

## **L3b:** Equity-Price Lattices Under Real-World and Risk-Neutral Measures

CHEME 5660 Cornell University

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# Objectives for Today

Today we replace continuous price uncertainty with a finite lattice, derive its node distribution, and ask which probability law answers which question.

## Today's concepts

- **Build a binomial lattice:** derive node prices and node probabilities from the up factor, down factor, horizon, and transition probability.
- **Separate probability measures:** explain why real-world probabilities support forecasting while risk-neutral probabilities support valuation.
- **Test model purpose:** evaluate probability of profit under the real-world measure and identify stylized facts excluded by fixed parameters.

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Companion notes: [L3b lecture notes and notebooks](#)

# Lectures and Examples

**Lecture:** CHEME-5660-L3b-Lecture-LatticeModels-Equity-Price-Fall-2026.ipynb

**Example:**

CHEME-5660-L3b-LatticeModelSharePrice-RWPM-Example-Fall-2026.ipynb

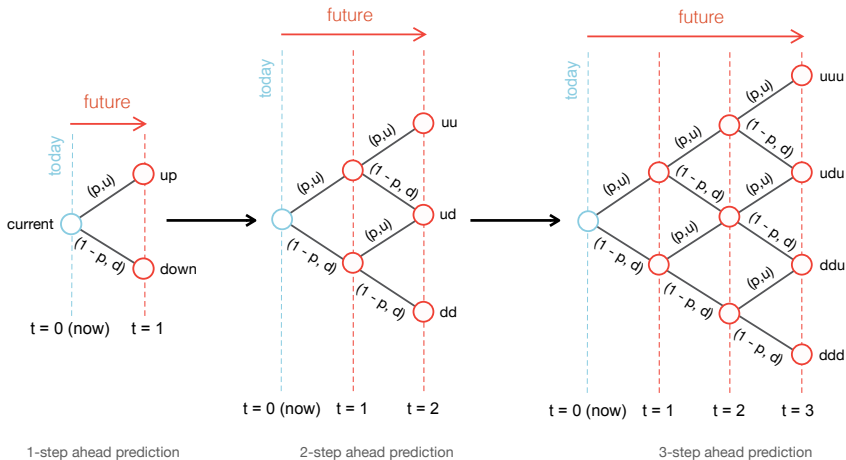
The example checks a textbook tree, estimates asymmetric move factors and a physical up probability from historical data, populates a recombining lattice, and compares its prediction bands with observed prices.

The factors are **not** forced to satisfy  $d = 1/u$ . That restriction is unnecessary for recombination.

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**Link:** <https://github.com/varnerlab/CHEME-5660-CourseRepository-Fall-2026.git>

# A Recombining Price Lattice



# Binomial Lattice Model

Let  $S_0 > 0$  be today's ex-dividend share price. Over each step of length  $\Delta t > 0$  years, the price is multiplied by a constant **up factor**  $u > 0$  or a constant **down factor**  $d > 0$ , with  $d < u$ .

Under a real-world up probability  $p \in [0, 1]$ , the one-step price is:

$$S_1 = \begin{cases} S_0 u, & \text{with probability } p, \\ S_0 d, & \text{with probability } 1 - p. \end{cases}$$

The fixed-parameter model assumes the same  $u$ ,  $d$ , and  $p$  at every node and independent move draws across steps. Level  $t$  contains  $t + 1$  price states.

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**Purpose:** the lattice replaces a continuum of possible prices with a finite map whose probabilities and payoffs can be computed exactly.

# Terminal Prices and Their Distribution

Let  $N$  be the number of independent steps, and let  $K_N$  count the up moves. Under the real-world law,  $K_N \sim \text{Binomial}(N, p)$ . Conditional on  $K_N = k$ , where  $k \in \{0, \dots, N\}$ , the terminal price is:

$$\underbrace{S_{N,k} = S_0 u^k d^{N-k}}_{\text{terminal price}}.$$

There are  $\binom{N}{k}$  paths with that count, and every such path has probability  $p^k(1-p)^{N-k}$ . The node probability is therefore:

$$\underbrace{\mathbb{P}(K_N = k) = \binom{N}{k} p^k (1-p)^{N-k}}_{\text{probability of node } (N,k)}.$$

The expected one-step gross return under this law is  $pu + (1-p)d$ .

# Paths Grow Exponentially, Nodes Grow Linearly

After  $N$  binary moves, the unmerged path tree contains  $2^N$  paths to the terminal level. The recombining lattice contains only  $N + 1$  terminal price nodes.

## Why the compression is exact

- Every path with  $k$  up moves and  $N - k$  down moves reaches  $S_0 u^k d^{N-k}$ .
- The binomial coefficient  $\binom{N}{k}$  records how many path orders merge into that node.
- Recombination preserves the terminal distribution while replacing path enumeration with one state per up-move count.

This state compression is what makes backward lattice calculations tractable.

# Real-World Parameter Estimation

Let  $g_2, \dots, g_T$  be historical growth rates sampled at interval  $\Delta t > 0$  years. Initialize an up-factor set  $U$  and a down-factor set  $D$ .

1. For every observation, place  $e^{g_t \Delta t}$  in  $U$  when  $g_t > 0$ , in  $D$  when  $g_t < 0$ , and skip the factor when  $g_t = 0$ .
2. Estimate the conditional move magnitudes:

$$\hat{u} = \frac{1}{|U|} \sum_{v \in U} v, \quad \hat{d} = \frac{1}{|D|} \sum_{v \in D} v.$$

3. Estimate the up probability by  $\hat{p} = |U|/(T - 1)$ , so zero-change days remain in the denominator.

The sample must contain at least one up move and one down move.

# Historical Estimates Are Model Choices

The procedure turns historical observations into a **real-world forecast law**. It does not make the parameters timeless or uniquely correct.

- Averaging positive and negative move factors compresses a continuous return distribution into two representative outcomes.
- Constant  $u$ ,  $d$ , and  $p$  discard serial dependence, changing volatility, and tail structure.
- Estimates inherit sampling error, the chosen observation window, data adjustments, and regime dependence.
- Validation must compare the entire out-of-sample distribution and path behavior with observations, not only the mean path.

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**Interpretation:**  $\hat{p}$  is an estimated physical frequency, not a risk-neutral pricing probability.

# Replacement, Recombination, and Reciprocal Moves

Three statements describe different features of the model.

- **With replacement:** every step draws again from the same move set  $\{u, d\}$ . An up move does not remove the possibility of another up move.
- **Recombining:** move order does not change the final price because multiplication commutes:

$$\underbrace{S_0 u d}_{\text{up then down}} = \underbrace{S_0 d u}_{\text{down then up}} .$$

- **Reciprocal moves:**  $d = 1/u$ , or  $ud = 1$ , makes up and down increments symmetric in log space. It is an additional parameter choice.

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**Vocabulary:** the unmerged path structure is a tree. After equal-price states merge, the lattice is a directed acyclic graph in the strict graph-theoretic sense.

## Example: Asymmetric Moves Still Recombine

The companion notebook first checks  $S_0 = 20$ ,  $u = 1.1$ ,  $d = 0.9$ , and  $p = 0.6523$ . These factors are asymmetric because:

$$d = 0.9 \neq \frac{1}{1.1} \approx 0.9091, \quad ud = 0.99 \neq 1.$$

The two mixed paths still reach one shared node:

$$\underbrace{20(1.1)(0.9)}_{\text{up then down}} = \underbrace{20(0.9)(1.1)}_{\text{down then up}} = 19.8.$$

Each ordered path has probability  $p(1 - p)$ , so the recombined middle node has probability  $2p(1 - p)$ . Reciprocal factors are unnecessary for both the shared price and the binomial node probability.

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**Notebook:** [CHEME-5660-L3b-LatticeModelSharePrice-RWPM-Example-Fall-2026.ipynb](#)

# Two Measures, Two Questions

The symbol attached to “probability of up” must identify its purpose.

## Real-world $\mathbb{P}$

- Up probability  $p$  is estimated or modeled from data.
- Used for forecasts, scenarios, risk, and probability of profit.
- Tested against out-of-sample distributions and paths.

## Risk-neutral $\mathbb{Q}$

- Up probability  $q$  is implied by  $u$ ,  $d$ , and the risk-free return.
- Used for no-arbitrage pricing and hedging.
- Tested by reproducing traded prices and excluding arbitrage.

A pricing probability is generally **not** an empirical frequency.

## Risk-Neutral Probability Enforces No Arbitrage

Let  $R_f > 0$  be the one-step risk-free gross return. For current equity price  $S_0$  and next-step prices  $S_0u$  and  $S_0d$ , absence of arbitrage requires:

$$d < R_f < u.$$

The unique risk-neutral up probability that makes the discounted share price a martingale is:

$$S_0 = \frac{qS_0u + (1 - q)S_0d}{R_f}, \quad q = \underbrace{\frac{R_f - d}{u - d}}_{\text{risk-neutral probability}}.$$

For a claim paying  $H_u$  in the up state and  $H_d$  in the down state, the same replication relation gives:

$$V_0 = \frac{qH_u + (1 - q)H_d}{R_f}.$$

This is an expectation under  $\mathbb{Q}$ , not a forecast of the realized payoff.

## Probability of Profit Belongs to the Real-World Measure

Let  $C_0 > 0$  be a strategy's initial cost,  $W_{N,k}$  its terminal wealth at node  $(N, k)$ , and  $R_b > 0$  a benchmark gross return per step. Under  $K_N \sim \text{Binomial}(N, p)$ , define profit by  $W_{N,K_N} > C_0 R_b^N$ . Its probability is:

$$\text{POP}_{\mathbb{P}} = \sum_{k=0}^N \mathbf{1}_{\{W_{N,k} > C_0 R_b^N\}} \binom{N}{k} p^k (1-p)^{N-k}.$$

POP depends on the forecast law, horizon, benchmark, costs, and profit definition. It is not expected profit and does not report loss severity.

Substituting risk-neutral  $q$  would answer a pricing-measure question, not the empirical probability that the strategy succeeds.

# Stylized Facts for Binomial Lattice Models

The lattice is our first model of equity-price dynamics. L3a gives three empirical requirements against which to test it.

- **Heavy tails:** large return magnitudes occur more often than a Gaussian model predicts.
- **Weak linear autocorrelation:** signed one-step returns usually carry little serial linear dependence.
- **Volatility clustering:** large movement magnitudes occur near other large movements, and calm observations near calm observations.

The fixed lattice matches the second pattern by assumption, but fails the first and third.

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**Source:** [Cont \(2001\)](#)

# Single-Step Growth Is a Bernoulli Transform

Let  $X_j$  equal one for an up move and zero for a down move, with  $X_j \stackrel{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$ . The one-step price ratio is:

$$\frac{S_j}{S_{j-1}} = u^{X_j} d^{1-X_j} = d \left( \frac{u}{d} \right)^{X_j}.$$

Let  $g_j$  be the continuously compounded growth rate over  $\Delta t > 0$  years. Taking logarithms gives:

$$g_j = \underbrace{\frac{1}{\Delta t} \left[ \log d + X_j \log \left( \frac{u}{d} \right) \right]}_{\text{two-point growth distribution}}.$$

The lattice growth rate is therefore an affine transformation of a Bernoulli variable and takes only two possible values.

# One-Step Mean and Variance

Let  $g_j$  be the two-point growth rate, with constant  $u > d > 0$ , probability  $p \in [0, 1]$ , and step length  $\Delta t > 0$ . Its expected value is:

$$\mathbb{E}[g_j] = \underbrace{\frac{p \log u + (1 - p) \log d}{\Delta t}}_{\text{mean growth rate}}.$$

Its variance is:

$$\text{Var}(g_j) = \underbrace{\frac{p(1 - p)}{(\Delta t)^2} \left[ \log \left( \frac{u}{d} \right) \right]^2}_{\text{fixed one-step variance}}.$$

The variance is positive when  $0 < p < 1$  and  $u \neq d$ . It vanishes at a degenerate probability or when the two moves coincide.

# Multi-Step Growth Is a Binomial Transform

Let  $K_t \sim \text{Binomial}(t, p)$  count the up moves in  $t$  steps. The cumulative growth quantity  $g_t = (1/\Delta t) \log(S_t/S_0)$  is:

$$g_t = \underbrace{\frac{K_t \log(u/d) + t \log d}{\Delta t}}_{\text{binomial transform}}.$$

Because  $\mathbb{E}[K_t] = tp$  and  $\text{Var}(K_t) = tp(1-p)$ :

$$\mathbb{E}[g_t] = \frac{t}{\Delta t} [p \log u + (1-p) \log d],$$

$$\text{Var}(g_t) = \frac{tp(1-p)}{(\Delta t)^2} \left[ \log \left( \frac{u}{d} \right) \right]^2.$$

The corresponding cumulative log return is  $r_t = (\Delta t)g_t = \log(S_t/S_0)$ .

# Heavy Tails

The fixed binomial lattice does not generate heavy-tailed growth.

- The one-step growth rate takes only two bounded values:  $\log u/\Delta t$  and  $\log d/\Delta t$ .
- The multi-step growth quantity is an affine transformation of the binomial count  $K_t$ .
- When  $0 < p < 1$  and  $u \neq d$ , the standardized binomial count converges to a Gaussian distribution as  $t$  grows.

The model therefore predicts extreme moves less often than the heavy-tailed empirical distributions examined in L3a.

# Vanishing Autocorrelation

Let  $g_j = b + aX_j$ , where  $a = \log(u/d)/\Delta t$ ,  $b = \log d/\Delta t$ , and the Bernoulli moves  $X_j$  are independent. For every nonzero lag  $\tau$ :

$$\text{Cov}(g_j, g_{j+\tau}) = a^2 \text{Cov}(X_j, X_{j+\tau}) = 0.$$

Provided  $0 < p < 1$  and  $u \neq d$ , the one-step variance is positive, so:

$$\underbrace{\rho_g(\tau) = 0}_{\text{one-step autocorrelation}} \quad (\tau \geq 1).$$

The lattice matches weak signed-return autocorrelation by construction. It does not discover that pattern from data: independence is an input assumption.

## Nested Windows Create Correlation

Let  $g_t = aK_t + bt$  be cumulative growth from step 0 through step  $t$ . The windows  $0 \rightarrow t$  and  $0 \rightarrow t + \tau$  share their first  $t$  moves, so their covariance is not zero:

$$\text{Cov}(g_t, g_{t+\tau}) = a^2 tp(1 - p).$$

Using the two variances gives:

$$\underbrace{\rho(\tau)}_{\text{nested-window correlation}} = \sqrt{\frac{t}{t + \tau}} \quad .$$

This correlation is mechanical overlap, not predictability in one-step moves. Nonoverlapping  $t$ -step windows have zero correlation under the i.i.d. assumption.

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**Careful:** always state whether a return series uses overlapping or nonoverlapping windows before interpreting its autocorrelation.

# Volatility Clustering

The fixed binomial lattice does not generate volatility clustering.

The one-step conditional variance is the same at every node and time:

$$\text{Var}(g_j \mid \mathcal{F}_{j-1}) = \frac{p(1-p)}{(\Delta t)^2} \left[ \log \left( \frac{u}{d} \right) \right]^2.$$

Constant  $u$ ,  $d$ , and  $p$  create constant conditional volatility. The model cannot produce persistent high- and low-volatility regimes unless those parameters or the move law become state dependent.

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**Connection to L3a:** near-zero signed-return autocorrelation can coexist with strong dependence in squared returns, but the fixed lattice excludes that second dependence.

# What the Fixed Lattice Cannot Reproduce

The lattice is valuable because its simplifications are explicit.

- It captures positive prices, a complete finite distribution, weak one-step autocorrelation, and tractable backward valuation.
- It excludes continuous move sizes, heavy tails, volatility clustering, regime changes, and parameter uncertainty.
- A prediction band can have acceptable average coverage and still fail during exactly the extreme or high-volatility periods that matter most.

Use a lattice to expose states, trading rules, and replication. Use richer innovations or state-dependent parameters when the question depends on rare moves or dynamic volatility.

# Optional Advanced Material

## **Recombination and computational cost**    advanced/recombination/

Compare the full binary path tree with its recombined lattice. Verify that asymmetric factors still recombine, count the saved nodes, and identify when time-dependent move factors prevent paths from merging.

## **Bootstrap calibration uncertainty**    advanced/calibration/

Resample historical growth rates, re-estimate  $(u, d, p)$ , and carry the resulting uncertainty into terminal-price and probability-of-profit calculations. Compare independent, block-bootstrap, and holdout results.

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**Notebooks:** [lectures/week-3/L3b/advanced](#), also linked from the Optional Advanced Material section of the lecture notebook.

# Summary

This lecture built a recombining equity lattice, separated forecasting from pricing measures, and tested the fixed model against empirical return patterns.

## Key takeaways

- **Recombination is algebraic.** A node depends on move counts because multiplication commutes. The constraint  $ud = 1$  is optional.
- **The measure follows the question.** Real-world  $p$  supports forecasts and POP, while risk-neutral  $q$  enforces no-arbitrage value.
- **Simple models have visible limits.** The fixed lattice matches weak one-step autocorrelation but misses heavy tails and volatility clustering.

The next step is to turn lattice states into explicit trading and valuation rules.

That's a wrap. Which lattice assumption would you relax first, and what empirical pattern should the extension recover?